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[illegible]

exactly 80 mm: _____

4 FIRST SOLVE

4 PUT THE YELLOW CROSS yellow edges only – ignore the corners



Dot no yellow edge at all
FRURU'U'



Hook two at a right angle: hold them BACK and LEFT
FRURU'



Line hold the bar left-to-right
FRURU'U'

One algorithm, up to three times: dot to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn it so at least two top edges match the center under them. This algorithm is called **Sune**; you use it on every card after this one.



two matched, side by side hold them BACK and LEFT
RU'R U' RU2'



two matched, opposite run it once from any hold, then look again
RU'R U' RU2'

6 PUT THE YELLOW CORNERS HOME



one corner home hold it FRONT-LEFT
RU'L U' RU'L



no corner home run it once from any hold, then look again
RU'L U' RU'L

This cycles the other three corners into place. It also twists the yellow face you just built – expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS



yellow faces RIGHT
R'D'R D' x2



yellow faces FRONT
D'R'D R x2

Between corners run **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look dented while you do this. Keep going – the last top turn brings it back.

LAST-LAYER ORDER On CARD: yellow cross, match edges, place corners, twist corners. CARD 2 replaces this order permanently – it is the only thing on this card that gets retired.

UNLUCKY CARD 2 – five scrambles in a row with this card face down. **5 MASTER:** those five under 3x0. My target: **NEXT: 2-3 TWO-LOOK**

FAST SOLVE

4 YELLOW CROSS

yellow edges only – ignore the corners



Do not yellow edge at all
FRUR'U'F'

Hook two at a right angle: hold them BACK and LEFT
FRUR'U'F'

Line hold the bar left-to-right
FRUR'U'F'

One algorithm, up to three times: do to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn it so at least two top edges match the center under them. This algorithm is called **Sune**; you use it in every card after this one.



two matched, side by side hold them BACK and LEFT
R'U'R'U' R2U'R'

look again

two matched, opposite run ONLY from any hold, then
R'U'R'U' R2U'R'

6 PUT THE YELLOW CORNERS HOME



one corner home hold it FRONT-LEFT
R'U'L'U' R'U'L'



no corner home run it once from any hold, then look again
R'U'L'U' R'U'L'

This cycles the other three corners into place. It also twists the yellow face you just built – expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS



yellow faces RIGHT
R'D'R'D x2



yellow faces FRONT
D'R'D'R x2

Between corners turn ONLY the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

LAST-LAYER ORDER On CARD 2: yellow cross, match edges, place corners, twist corners. CARD 2 replaces this order permanently – it is the only thing on this card that gets retired.

UNLOCK CARD 2 – five scrambles in a row with this card face down. > MASTER: those five under 3:00. My target: ~. NEXT: 2/3 TWO-LOOK

2-LOOK PLL CORNERS

headlights = two matching corners on one face



T headlights on ONE face – hold that face LEFT

RUR'U' R'F **RUR'U' R'UR'**



Y no headlights – the two swapping corners sit on the back face

FRU'R' R'UR'F **RUR'U' R'FR'**

PLL EDGES

corners home, top still not solved



Ua the front edge travels LEFT

RZU **RUR'U' R'U' R'UR'**



Ua the front edge travels RIGHT

MZU **MU'U' M'U'U' M'**



H both pairs swap straight across

MZU'MZ U'U' MZU'MZ



Z each pair swaps with its neighbour

M'U' M'U'ZU'MZ' M'U'U' MZU

START HERE

two new algorithms finish any last layer
 Yellow cross: **F R U R' U' F** until the cross appears – at most 3.
 Yellow face: Same, read again, repeat – at most 3 (see the badges).
 Corners: F-form. No headlights to hold left? Run it twice.
 Edges: Ub. No solved edge to put at the back? Run it twice.
 Each case on the front replicates one repeat with one run. Three a week; you can solve the whole time.

HVSZ

Both solve headlights on all four faces. The edge between them decides if it belongs to the opposite face = R, to a neighbour = Z.

STUCK?

Corners home but nothing solves? Turn the top face once and read again – that is often the whole fix.
 One case has beaten you to the punch? Park it, use its fallback above, come back next time.
 A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and repeat it.

WORDS

CLK = make the whole top face yellow - PLL = slide the top pieces home - headlight = two matching corners on one face - **AUF** = the free top turn that lines a case up

THE NEW ORDER: yellow2M, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas; it moves corners but wrecks the yellow face, and here the yellow face is finished first.

UNDER CARD 3 – different last layers in a run, each finished in exactly two algorithms, case named out loud first. A repeat does not count, o MASTER: solves averaging under 1:00. My target: 1:00. NEXT: 3-ONE LOOK PLL

2-LOOK PLL CORNERS headlits = two matching corners on one face  T headlits on ONE face = hold that face LEFT RUR'U' R'F R2UR'U' RUF'	START HERE Two new algorithms finish any last layer Yellow cross: F R U R' U' F until the cross appears – at most 3. Yellow face: Same, read again, repeat – at most 3 (see the badges). Corners: F-Term. No headlits to hold left! Run it twice. Edges: U. No solved edge to put at the back3! Run it twice. Each case on the front replaces one repeat with one run. Three a week, you can solve the whole time.
2-LOOK PLL EDGES corners home, top still not solved  Y no headlits – the two swapping corners sit neutral FR'U'R' RUR'F R'UR'U' R'FR'	HVSZ Both show headlits on all four faces. The edge between them decides if it belongs to the opposite face = H, to a neighbour = Z.
Ua the front edge between LEFT R2U RUR'U' R'U' UR'  Ua the front edge between RIGHT M2U M2U' M'U'U2	STUCK! Corners home but nothing solves? Turn the top face once and read again – that is often the whole fix. One case has been solved in five times! Park it, use its fallback above, come back next week. A single piece looks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and reseat it.
H both pairs swap straight across M2U'M2 U2 M2U'M2  Z each pair swaps with its neighbour M'U' M'U'2M2U' M'U2 M2U	WORDS OLL = make the whole top face yellow - PLL = slide the top pieces home - headlits = two matching corners on one face - AUF = the free top turn that lines a case up

EDGES ONLY corners already home

 • **U**1b **B** solid: F.L.R. headlights; 3 edges counter-clockwise
M2U M'U2 M'U2M

• **U**1b **B** solid: F.L.R. headlights; 3 edges clockwise
R2U R'U'R' R'U' R'UR'

• **H** all 4 headlights; opposite edges swap
M2U'M2 U2 M2U'M2

• **Z** all 4 headlights; adjacent edges swap
M'U' M2U'M2U' M'U2 M2U

DIAGONAL CORNER SWAP no headlights anywhere

 • **Y** twin: F-left R-back
FR'U'R'U' R'U'R'F' R'U'R'F'

 • **Y** pairs: R-left L-front
Y'R'U'R' Y'F' R2U'R'U' R'FR'

 • **N** naive: B-left F-right L-front R-back
R'U'R'U' R'U'R'U' R'U'R'F' R'U'R'F'

 • **N** naive: B-right F-left L-back R-front
R'U'R'U'R' F'U'F' R'U'R' F'F' R'U'R'

HOW TO LOOK

- Count the headlight faces. One = right of this card. Four = this column. None = the column on the right
- Only then read the edges to pick the exact case.
- Line the case up as shown; run, turn the top again to finish – that last free turn is the AUF.
- Done! You're yours from Card 2.

WORDS

PLL = slide the top pieces home = headlights = 2 matching corners on the free top turn that then cause up-adjacent corner swap = the swapping corners share an edge = diagonal corner swap = the swapping corners are opposite

Card 2 gives you 19/72 last layers in one look. This card gives 7/72. Next: the other 7 top-face cases, and the first two twenty-ones as pairs – drilled with a cube in the app: cubeplus-six-vertex/app practice

Done! all twenty-one named correctly, twenty-one in a row, no two separate days, on two separate days; in 30 seconds; PLL under 0:30. My target: 100%

EDGE ONLY

corners already home



• **U** **B** solid: F L R headlights: 3 edges counter-clockwise
M2U MU2 M'UM2

• **U** **B** solid: F L R headlights: 3 edges clockwise
R2U R'UR' R'U' R'UR'

• **H** all 4 headlights: opposite edges swap
M2U'M2 U2 M2U'M2

• **Z** all 4 headlights: adjacent edges swap
M2U'M2M2U' M'U2 M2U

DIAGONAL CORNER SWAP

no headlights anywhere



• **V** pairs: F-left 3 back
FR'UR'U' R'UR'F' R'UR'U' R'RF'

• **V** pairs: F-left L-front
R'UR'U' y R'F' R2UR'U' R'RF'

• **N** pairs: B-left F-right L-front 3 back
R'UR'U' R'UR'F' R'UR'U' R'F' R'UR'U' U2R'UR'

• **N** pairs: B-right F-left L-back B-front
R'U' UR'UR' F'UF' R'UR' F'RF' R'UR'

• **E** nothing matches
X' L'UL'D' L'UL'D' L'UL'D' L'UL'D'

HOW TO LOOK

- Count the headlight faces. One - front of this card. Four = this column. None = the column on the right
- Only then read the edges to pick the exact case.
- Line the case up as drawn, run it, turn the top again to finish – that last free turn is the AUF.
- Now you're from Card 2.

WORDS

PLL = slide the top pieces home - headlights = 2 matching corners on one face = 1 free top turn that makes a case up
adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners are opposite
Card 2 gives you 1/722 last layers in one look. This card gives 7/172.
Note: the other 57 face-matched cases, and the first two layers on pairs – drilled with a cube in the app: cubeheads3x3verred.app/3x3solve
DONE
My target: all twenty-one named corners, twenty-one in a row, no card in hand, on two separate days; on two separate days; on 3 practices

THE LADDER

1/3 FIRST SLOPE -> algorithms - unlock: five-saves, card face down

2/3 TWO-LOOK -> algorithms - unlock: fifteen last letters in exactly two algorithms

3/3 TWO-LOOK PLL - +15: done! all 21 named, twice on separate days

After that: full OLL, F2L, cross planning, lock-ahead - in the app, on card stock

WHY THERE ARE ONLY THREE

Card is good at a clockwise set of cases you tell apart by sight. It is the only one you drill. F2L is the only lock-ahead are drills with no finite case list. Full OLL is 75 cases, 22 of which differ only in a divide fifth of a millimetre wide at this size. All of them are in the app, with a real cube and randomised set. This set stops when the medium stop

PRINT AND VERSION

EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time: if one were wrong the build would fail. Recognition wording is generated from the same permutation the diagram is drawn from.

Set v1.0 - replace any single card at cubepu3h-s1-buy/app/print - print at 100mm / Actual size, never 'Fit to page'

Not a tire, no unlock, no name. Use it when you've a 4x4.

WORDS

OLL = make the whole top face yellow - PLL = slide the top corner - none - Sune = the one-yellow-corner algorithm, use on every card after - seiyu move = the R U F' R' U' F' sequence - slice move = the R F R' F' - headlight - two matching corners on one side - a slice move - the R U' R' U' sequence - slice move - the R F = the free top turn takes the case up parity = a case a 3x3 cannot have: edge pair + two edges glued into one on a 4x4 or 5x5 - the first two layers, solved as pairs: lock ahead = watching the next piece while your hands finish this one - adjacent corner swap = the swapping corners - slice move - diagonal corner swap = the swapping corners sit opposite

THE LADDER

A1 1/3 FIRST SLOPE = 9 algorithms - unlock: five solves, card face down.

A2 2 FOUR-LOOK = 10 algorithms - unlock: fifteen last layers in exactly two faces.

A3 3 SIX-LOOK PL1 +15: done in all 21 named, twice on separate days.

After that: full OLL, F2L, cross planning, look-ahead – in the app, on card stock.

WHY THERE ARE ONLY THREE

A CARD IS good at a closest set of cases you tell apart by sight. It is a diver's depth drill. The ladder's look-ahead are drills with no finite case limit. Full OLL is 75 cases, 22 of which differ only in a diver's fifth of a millimetre wide at this size. All of them are in the app, with a real cube and ranked sets. This sets stops where the medium stop.

PRINT AND VERSION

EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were wrong the build would fail. Recognition words is generated from the same programming diagram as drawn from. Set v1.0, repeat any single card at cubepush-site/buycard/app/print - print at 100% / Actual size, never Fit to page.

Not a turn, no unlock, no record. Use it when ever you buy a toy4.

WORDS

OLL = make the whole top face yellow - PLL = slide the top pieces home - Slice = the one-yellow-corner algorithm, used on every card after the very move in the ETRUF trigger - edge-matching - the R F R F trigger - headlights = two matching corners on one side - slice-headlights = two hands ready to swap - the R F = the free top turn that lines a case up - parity = a case a 3x3 cannot have - edge pairs = two edges glued into one pair - solved pairs = two edges, solved as pairs - look-ahead = watching the next piece while your hands finish this one - adjacent corner swap = the swapping corners - diagonal corner swap = the swapping corners sit opposite

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numer